Md Basim Uddin Ahmed

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Software engineer and MSc Computer Science graduate with three years of professional Java backend experience and research in LLM agents, software security, and automated testing. Built Spring Boot REST APIs, PostgreSQL workflows, CI/CD integrations, and Python developer tools; developed React interfaces through personal projects.

Best Dataset & Benchmark Paper, AIware 2026 - SecVulEval.

Technical Skills

Languages	Java, Python, JavaScript, SQL, Bash, Groovy
Backend and data	Spring Boot, Spring Framework, REST APIs, PostgreSQL, Hibernate, Redis, MySQL, SQLAlchemy
Frontend	React, JSX, Astro, Tailwind CSS, i18next / react-i18next
LLM agents	OpenAI SDK, Claude Code, Agent Skills, LangGraph, RAG, OpenSearch, FastMCP, LiteLLM, OpenCode
Observability	Opik, Langfuse, OpenTelemetry, forward-proxy tracing, trace/span analysis, root-cause analysis
ML and analysis	PyTorch, OpenCLIP, Transformers, torchvision, NumPy, Pandas, Matplotlib, OpenCV
Testing and delivery	JUnit, debugging, Git, GitHub Actions, Jenkins, Gradle, Maven, Docker, Linux, Jira, Confluence
Research methods	Benchmark construction, LLM evaluation, static analysis, metamorphic testing, ablations, experiment design, failure analysis

Education

York University — Sep 2024 - Aug 2026
MSc in Computer Science — Toronto, ON, Canada
Thesis: **SecVulEval: Benchmarking LLMs for Real-World C/C++ Vulnerability Detection**.
Supervisor: Dr. Song Wang.

Shahjalal University of Science and Technology — Feb 2017 - Jul 2021
BSc (Engg.) in Computer Science and Engineering — Sylhet, Bangladesh
CGPA: **3.73/4.00**. Bachelor's research: Bengali topic modeling and news-trend analysis.

Professional Experience

Huawei Research Institute — Aug 2025 - Sep 2026
Research & Development Intern - University Collaboration — Canada

- Developed **Python** components for repo-wide test generation from software requirements and source code.
- Built agent workflows with Claude Code, OpenAI SDK, and OpenSearch/RAG; evaluated retrieval and code maps.
- Built a forward-proxy framework to capture session-linked prompts, responses, tool calls, and execution traces.
- Analyzed Opik/Langfuse traces using OpenTelemetry; assessed instruction violations, tool misuse, and failure causes.
- Defined evaluation labels and documented findings; collaborated remotely and on-site with the Beijing research team.

Tech stack: Python, OpenAI SDK, Claude Code, OpenSearch, RAG, Opik, Langfuse, OpenTelemetry, Bash, Linux.

Dynamic Solution Innovators Ltd. — Jul 2021 - Jul 2024
Associate Software Engineer — Dhaka, Bangladesh

- Developed **Java/Spring Boot REST APIs** and PostgreSQL features for HEMS, a national examination platform.
- Designed backend services and schemas from client requirements; reviewed code and fixed defects.
- Built marks aggregation, grading, and reporting workflows for approximately **30,000 students annually**.

Dynamic Solution Innovators Ltd. (continued)

- Integrated **two payment gateways** processing about **BDT 4 million in annual transactions**.
- Built JWT-authenticated REST APIs for mobile clients.

Tech stack: Java, Spring Boot, Spring Framework, PostgreSQL, Hibernate, Redis, REST APIs, Git, Linux.

Synopsys Inc. Aug 2023 - Jul 2024
Software Engineer (External) - Assignment through DSi Remote, USA

- Contributed **Java**, Groovy, and Gradle changes to Black Duck Detect, a software composition analysis tool.
- Helped migrate CI/CD to GitHub Actions; fixed two defects in job scheduling and artifact versioning.
- Migrated Universal Analytics to GA4, restoring the product reporting dashboard.

Tech stack: Java, Groovy, Gradle, Jenkins, GitHub Actions, Google Analytics 4.

Research Experience

SecVulEval - Context-Aware C/C++ Vulnerability Benchmark 2024 - 2026
Master's research, York University | AIware 2026

- Built a benchmark of **25,440 functions and 5,867 CVEs** from **707 projects**, covering **145 CWE types**.
- Curated vulnerable/fixed pairs, statement-level labels, and external context; released the dataset on Hugging Face.
- Built an agentic detection/validation pipeline and evaluated **five LLMs** at function and statement levels.
- Validated 1,000 context samples; ablation showed a **4.96-point F1 drop** for GPT-4.1 without the Context Agent.
- Received the **Best Dataset & Benchmark Paper award at AIware 2026**.

Tech stack: Python, OpenAI SDK, Transformers, benchmark construction, context-aware agents, empirical evaluation.

StaAgent - Testing Java Static Analyzers Mar 2025 - Jul 2025
Research project, York University | ACM TOSEM article, accepted 2026

- Developed an LLM-assisted metamorphic-testing framework for Java static analyzers.
- Implemented four agents for seed generation, validation, mutation, and analyzer-consistency testing.
- Identified **42 faulty rule implementations in SpotBugs, Error Prone, and PMD**; reported findings to developers.

Tech stack: Python, Java, OpenAI SDK, Transformers, SpotBugs, Error Prone, PMD, metamorphic testing.

Publications

Md Basim Uddin Ahmed, Nima Shiri Harzevili, Jiho Shin, Hung Viet Pham, and Song Wang. "SecVulEval: Context-Aware Benchmarking of LLMs for Vulnerability Detection." *Proceedings of the 3rd ACM International Conference on AI-Powered Software (AIware)*, 2026. DOI: 10.1145/3805760.3814932. **Best Dataset & Benchmark Paper.**

Elijah Nnorom*, **Md Basim Uddin Ahmed***, Jiho Shin, Hung Viet Pham, and Song Wang. "StaAgent: An Agentic Framework for Testing Static Analyzers." *ACM Transactions on Software Engineering and Methodology (TOSEM)*, accepted, 2026.
**Equal Contribution*

Md. Basim Uddin Ahmed, Ananta Akash Podder, Mahruba Sharmin Chowdhury, and Mohammad Abdullah Al Mumin. "A Systematic Literature Review on English and Bangla Topic Modeling." *Journal of Computer Science*, 17(1), 1-18, 2021. DOI: 10.3844/jcssp.2021.1.18.

Projects

PairCheck - Image-Text Mismatch Detection Sep 2026
Multimodal machine learning

- Built an image-caption mismatch detector with frozen CLIP embeddings and a trainable PyTorch head.
- Built COCO image-level splits and **nine hard-negative types**; used balanced batches and validation selection.
- Evaluated the **OpenCLIP ViT-B/32 baseline** on SugarCrepe: **76.55% pairwise accuracy**.

Tech stack: Python, PyTorch, OpenCLIP, COCO, SugarCrepe.

LLM Agent Trace Analyzer 2026

Huawei R&D - Agent reliability

- Linked task requirements, tool calls, traces/spans, and generated outputs for agent failure analysis.
- Built planning, instruction-following, and tool-use assessments with resumable caching for LLM evaluations.

Tech stack: Python, Opik, Langfuse, OpenTelemetry, OpenAI SDK, LLM-based evaluation.

SaliencyMix Replication Mar 2025 - Apr 2025

Computer vision and deep learning

- Implemented SaliencyMix augmentation components, including saliency maps and salient-region mixing.
- Implemented **ResNet-18/50/101** in PyTorch and workflows for CIFAR-10, CIFAR-100, and ImageNet.

Tech stack: Python, PyTorch, torchvision, NumPy, OpenCV.

iambasim - Personal Portfolio Aug 2025

Frontend development

- Built a responsive Astro/React portfolio with reusable navigation, project modals, and dark mode.
- Added English/French localization with i18next; styled with Tailwind CSS and deployed on Netlify.

Tech stack: JavaScript, React, JSX, Astro, Tailwind CSS, i18next, Netlify.

DeObfus Email Jun 2024

Browser extension

- Built a rule-based browser extension that converts obfuscated email text into copy-ready addresses.
- Handled Firefox/Chromium API differences, permissions, and Manifest V3 compatibility.

Tech stack: JavaScript, Browser Extension APIs, Manifest V3.

HF ML App Analysis Nov 2023

Data analysis and software maintainability

- Collected and filtered applications built on top-downloaded Hugging Face classification/generation models.
- Extracted code-size metrics with Lizard; analyzed implementation-size and maintainability trends.

Tech stack: Python, Pandas, NumPy, Matplotlib, Lizard.

Teaching

York University - Electrical Engineering and Computer Science

Teaching Assistant Toronto, ON, Canada

Supported undergraduate students in computing courses.

Kendrobindu Academic Care Jan 2017 - Dec 2020

Tutor Brahmanbaria, Bangladesh

Taught physics and programming; prepared questions, invigilated tests, and graded student work.

Certifications

- **Neural Networks and Deep Learning** - Coursera / DeepLearning.AI.
- **Sequence Models** - Coursera / DeepLearning.AI.